

PSTAT 197B Interim Report

Singularity Solutions

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Introduction

Subsurface drilling operations are fundamental to industries based around constructing infrastructure such as energy production and geotechnical engineering. Understanding the geometry of boreholes is critical for evaluating structural integrity and predicting mechanical stresses to avoid important issues in the future. Accurate geometric representations of boreholes could definitely assist engineers in identifying irregularities prior to the actual installation of pipes.

Modern drilling simulations and sensing technologies often generate large three-dimensional point cloud datasets representing the surrounding rock formation and the borehole structure. These datasets can consist of millions of spatial points describing the geometry of the environment. The issue with these point cloud datasets is the raw data does not explicitly contain a surface representation of the borehole. Instead, the borehole wall must be inferred from scattered spatial measurements. This creates a very difficult computational problem because the data contains noise, irregular sampling density, and incomplete coverage of the borehole boundary.

The presence of this noise and measurement artifacts further complicates the analysis. Also the boreholes are not always perfectly cylindrical due to geological variability, drilling dynamics, and etc. Extracting a reliable surface representation from raw point cloud data requires robust computational methods capable of handling irregular geometry and noisy measurements.

This capstone project is of course conducted in collaboration with Cavator.ai, a company focused on improving drilling analytics through advanced computational tools. The primary objective of this project is to develop methods for identifying the boundary of a borehole from three-dimensional point cloud data and reconstructing a geometric representation of the borehole surface. By transforming raw point cloud measurements into an interpretable surface representation, the project aims to support downstream analyses of borehole geometry and structural behavior.

The remainder of this report is organized as follows. First, we define the primary problems addressed by this project. Next, we describe the datasets and computational methods used to analyze the point cloud data. We then present preliminary findings demonstrating boundary detection and cross-sectional analysis of the borehole structure. Finally, we discuss challenges encountered during the analysis and outline directions for future work.

Problems of Interest

Borehole Boundary Identification

The first problem is identifying which points in the dataset correspond to the borehole boundary. The raw point cloud contains measurements from both the borehole wall and surrounding geological structures which makes it necessary to develop methods that distinguish boundary points from interior or exterior observations.

Surface Reconstruction

After boundary points are identified, the next problem is reconstructing a smooth geometric surface representing the borehole wall. This involves transforming scattered boundary points into a continuous representation of the borehole geometry.

Geometric Characterization of the Borehole

The final problem involves analyzing the reconstructed surface to estimate geometric properties of the borehole, such as radius variation and deviations from ideal cylindrical shape. These measurements can provide insights into drilling performance and potential structural irregularities.

Materials and Methods

Dataset

The primary dataset used in this project is a three-dimensional (3D) point cloud representing the material surrounding a drilled borehole. The data was provided by Cavator.ai and consists of approximately 2.8 million spatial observations stored in VTK format. Each observation corresponds to a point in space described by Cartesian coordinates (x,y,z) .

Unlike many 3D datasets used in computer graphics, this dataset does not include explicit surface information such as mesh connectivity or labeled boundaries. Instead, it consists only of discrete points sampled from the surrounding material. As a result, the geometry of the borehole must be inferred from the spatial structure of the point cloud.

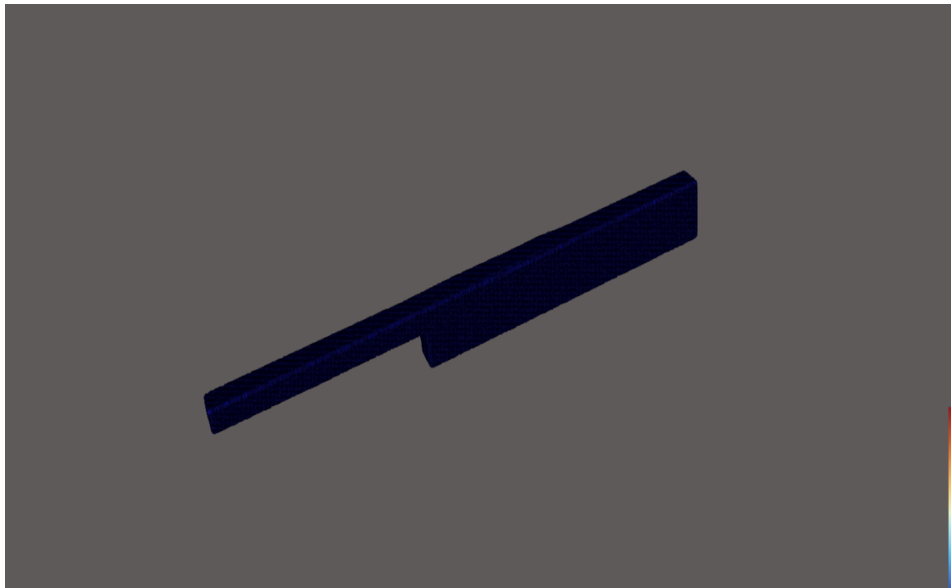


Fig. 1. High resolution point cloud data where each point represents a particle of displaced soil. In general, the surrounding material contains a high concentration of points, while the interior of the borehole contains relatively few points. This difference in point density provides an important signal that can be used to identify the borehole region.

However, the dataset also presents several challenges. Point density is not uniform throughout the scan, the borehole boundary is not always fully observed in every cross-section, and the data may contain noise or measurement artifacts. These properties require a reconstruction approach that is robust to incomplete or irregular observations.

Reconstruction Strategy

The overall goal of the project is to estimate the shape and position of the borehole from the raw point cloud data. Rather than attempting to reconstruct the entire three-dimensional surface at once, the approach used in this project analyzes the data through a series of two-dimensional cross-sections.

The point cloud is divided into thin slices along a chosen axis. Each slice is analyzed independently to identify the location and shape of the borehole cross-section. Once these cross-sections are estimated, they can be combined to reconstruct the borehole geometry along the length of the scan.

This slice-based strategy simplifies the reconstruction problem by reducing it to a sequence of two-dimensional detection tasks.

The main stages of the reconstruction pipeline are:

1. Extract cross-sectional slices of the point cloud.
2. Convert slice points into a smoothed density field.
3. Identify low-density regions corresponding to the borehole cavity.
4. Select candidate boundary points surrounding the cavity.
5. Estimate the borehole cross-section using an ellipse model.
6. Combine cross-sections to approximate the three-dimensional structure.

Each stage is described in more detail below.

Slice Extraction

The first step in the reconstruction process is to extract thin cross-sections of the point cloud. A slice is defined by selecting all points whose coordinate along the slicing axis lies within a small distance of a specified position.

For example, when slicing along the x-axis, a slice centered at position $\mathbf{x}_0$ includes all points satisfying

$|\mathbf{x}-\mathbf{x}_0| < \delta$ where δ determines the thickness of the slice.

The selected points are then projected into two dimensions by discarding the slicing coordinate. This produces a set of two-dimensional points representing a cross-section of the surrounding material and the borehole cavity.

Density Field Construction

Because the points in each slice are irregularly distributed, they are converted into a structured representation using a two-dimensional histogram. The histogram divides the slice into a grid and counts how many points fall into each grid cell.

This grid-based representation provides a coarse estimate of point density across the slice. To reduce noise and smooth out small fluctuations in density, a Gaussian filter is applied to the histogram. The result is a smoothed density field in which regions with many nearby points appear as high-density areas.

In this representation, the surrounding material forms dense regions, while the borehole interior appears as a low-density region.

Low-Density Region Detection

To identify potential borehole cavities, the density field is thresholded using a quantile-based rule. Grid cells with density values below a specified percentile are classified as low-density cells. This produces a binary mask that separates candidate cavity regions from the surrounding material.

However, thresholding alone may produce small spurious regions caused by noise or uneven sampling. To address this, morphological operations are applied to clean the binary mask. These operations remove small isolated regions and fill small gaps in the detected cavity.

Cavity Identification

The cleaned binary mask may contain multiple disconnected regions. Connected-component analysis is used to label each region and measure its geometric properties.

Regions that touch the outer boundary of the grid are discarded because they represent open areas outside the borehole rather than enclosed cavities. The remaining regions are evaluated using simple geometric criteria such as size and compactness.

The region that best satisfies these criteria is selected as the borehole cavity for that slice.

Boundary Point Selection

Once the cavity region has been identified, its boundary is extracted. The boundary represents the interface between the low-density cavity and the surrounding material.

Rather than fitting a geometric model directly to grid cells, the algorithm selects nearby points from the original point cloud that lie close to this boundary. These points form the candidate set used to estimate the shape of the borehole cross-section.

Ellipse-Based Cross-Section Estimation

The borehole cross-section is modeled using an ellipse fitted to the candidate boundary points. Ellipses provide a flexible representation that can capture circular, elliptical, or moderately irregular cross-sections.

The orientation of the ellipse is estimated using Principal Component Analysis (PCA). PCA identifies the directions along which the candidate points vary the most, which correspond to the major and minor axes of the ellipse.

$$\Sigma = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)(x_i - \mu)^T$$

Once these directions are determined, the semi-major and semi-minor axis lengths are estimated from the spread of the points along each direction. Together with the estimated center of the points, these parameters define an ellipse that approximates the borehole cross-section for the slice.

Preliminary Findings

The slice-based reconstruction pipeline successfully identifies candidate borehole regions in many cross-sections of the dataset. The density-based approach is effective because the borehole interior consistently appears as a region of reduced point density relative to the surrounding material.

In slices where the borehole boundary is clearly represented, the cavity detection algorithm produces well-defined interior regions, and the PCA-based ellipse fitting method provides a stable estimate of the cross-sectional geometry.

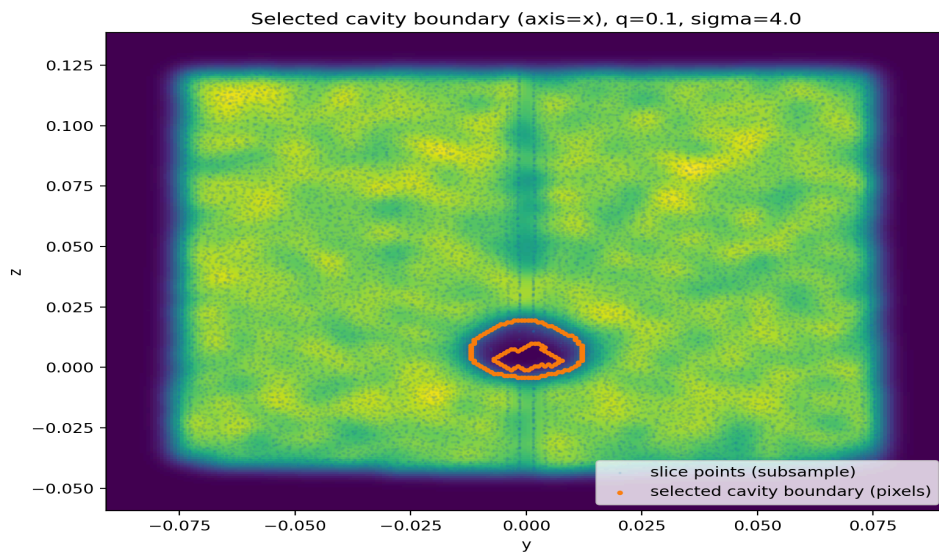


Fig. 2. Slice of the point cloud and the corresponding density field used to identify the borehole cavity.

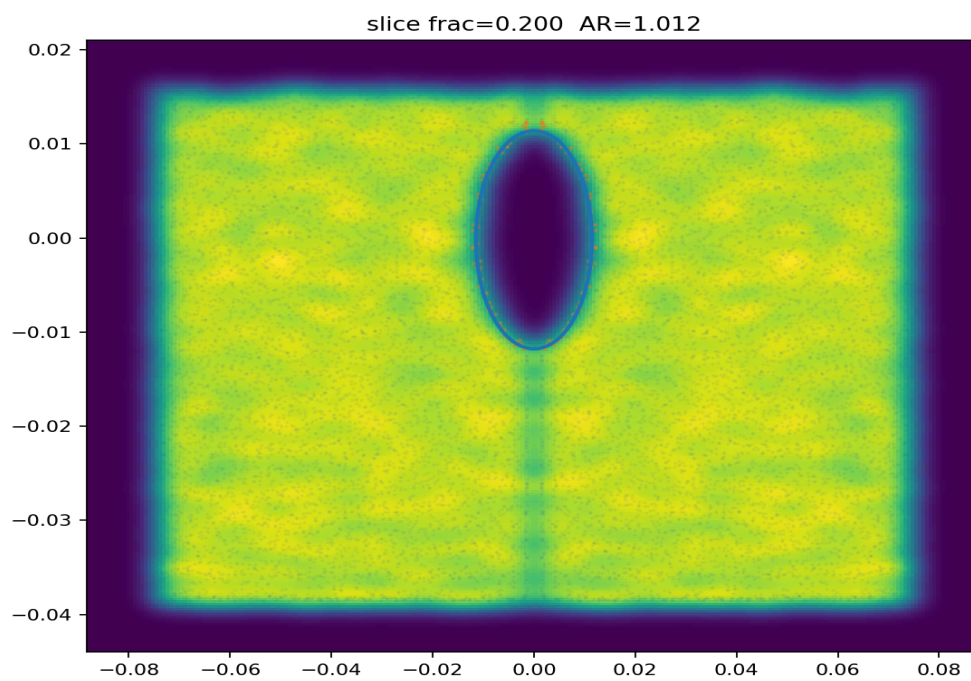


Fig. 3. Fitted ellipse overlaid on the candidate boundary points.

These results demonstrate that the pipeline is capable of identifying the borehole location and estimating its cross-sectional shape in many slices of the dataset.

However, the results are not yet fully consistent across all slices. In some cases, irregular sampling or noise causes the cavity detection step to produce ambiguous regions. In other cases, the borehole boundary is only partially observed, which can lead to unstable ellipse estimates.

Despite these challenges, the method provides a promising foundation for reconstructing the borehole geometry from the point cloud data.

Discussion

The preliminary results indicate that a slice-based density segmentation approach can successfully identify borehole cavities in unstructured point cloud data. By converting irregular point distributions into smoothed density fields, the method provides a reliable way to separate the borehole interior from the surrounding material.

The use of PCA-based ellipse fitting offers a simple and computationally efficient method for estimating cross-sectional geometry. Because PCA identifies the dominant directions of variation in the candidate boundary points, it can provide reasonable estimates even when the borehole boundary is only partially observed.

At the same time, several limitations remain. The density threshold used to identify cavity regions must be carefully chosen, as overly aggressive thresholds may remove valid boundary information, while overly permissive thresholds may include noise.

In addition, the borehole cross-section is not guaranteed to be perfectly elliptical. While the ellipse model provides a useful approximation, some slices may contain irregular shapes that are not well captured by this representation.

These challenges suggest that additional constraints or modeling approaches are necessary to improve the robustness of the reconstruction.

Future Work

Several directions for future work are planned for the next phase of the project.

First, the reconstruction pipeline will be extended to combine the cross-sectional ellipses into a continuous three-dimensional surface representation of the borehole. This will involve connecting adjacent slices to generate a surface mesh that approximates the borehole wall.

Second, the stability of the cavity detection step will be improved by refining the density thresholding procedure and incorporating additional geometric criteria for selecting candidate regions.

Third, alternative shape models may be explored to better capture irregular borehole geometries. For example, piecewise or non-elliptical representations may provide a more accurate description of certain cross-sections.

Finally, additional diagnostic visualizations will be developed to evaluate reconstruction quality and identify failure cases more effectively.

These improvements will hopefully produce an accurate representation of borehole geometry from the scanning data.